SQL Data Cleaning Pipeline

ETL TRANSFORMATION & VERIFICATION SCRIPT — BANK_COLLECTIONS_RAW

The SQL script below handles missing values, standardizes date formats, cleans whitespace, casts monetary metrics to numeric types, and creates the clean reference view.

```sql
-- 1. Create Cleaned Target Table
DROP TABLE IF EXISTS bank_collections_cleaned;
CREATE TABLE bank_collections_cleaned AS
SELECT
    TRIM(loan_id) AS loan_id,
    TRIM(account_id) AS account_id,
    CAST(payment_date AS TIMESTAMP) AS payment_date,
    COALESCE(CAST(amount_paid AS DECIMAL(12,2)), 0.00) AS amount_paid,
    CASE
        WHEN LOWER(TRIM(payment_status)) IN ('verified', 'approved', 'success') THEN 'Verified'
        WHEN LOWER(TRIM(payment_status)) IN ('pending', 'processing') THEN 'Pending'
        ELSE 'Rejected'
    END AS payment_status,
    COALESCE(CAST(reported_claim AS DECIMAL(12,2)), 0.00) AS reported_claim
FROM bank_collections_raw
WHERE loan_id IS NOT NULL;

-- 2. Audit Anomaly Records (Unverified or High-Variance Claims)
CREATE OR REPLACE VIEW vw_claims_audit AS
SELECT
    loan_id,
    account_id,
    amount_paid,
    reported_claim,
    (reported_claim - amount_paid) AS variance_amount
FROM bank_collections_cleaned
WHERE payment_status = 'Pending'
    OR (reported_claim - amount_paid) > 50000;

-- 3. Verify Cleaned Output Count
SELECT payment_status, COUNT(*) AS total_records, SUM(amount_paid) AS total_paid
FROM bank_collections_cleaned
GROUP BY payment_status;
```